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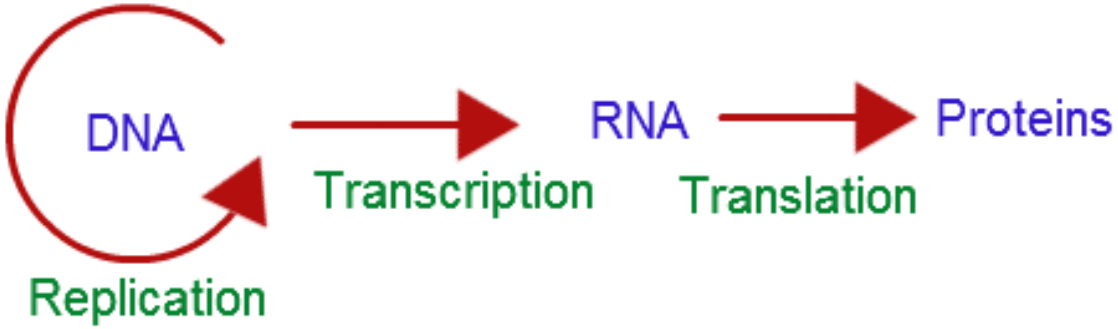
# Central Dogma of life

Transcription and Translation

# Central Dogma of life



- The information content of DNA is in the form of specific sequences of nucleotides
- The DNA inherited by an organism leads to specific traits by dictating the synthesis of proteins
- **Gene expression,** the process by which DNA directs protein synthesis, includes two stages: transcription and translation
- Genes specify proteins via transcription and translation
- The central dogma of biology describes the flow of information from gene sequence to protein product



# An Analogy



- When you get to the library, you search through the shelves until you find a book that has a good set of instructions for making a table.
- The library doesn't have a photocopier, but you came well-prepared and you whip out your pen and notebook and copy down (or 'transcribe') the instructions.
- Satisfied, you take your notes and head back home.
- At home, you have all the wood and tools to make the table.
- You follow the instructions and make the table, effectively 'translating' the written words into a table which you can use in your room.

# An Analogy



- Library = Nucleus
- Books in the library = DNA in the form of chromosome
- One chapter = One gene ( contains instruction for one type of protein)
- Notebook = Represents RNA, which is a smaller molecule that can move out of the nucleus into the cytoplasm, where proteins are made
- Home = Cytoplasm
- Table = One protein product
- Wood to make the table = amino acids for building proteins

# Basic Principles of Transcription and Translation



- RNA is the bridge between genes and the proteins for which they code
- **Transcription** is the synthesis of RNA using information in DNA
  - Transcription produces **messenger RNA (mRNA)**
- **Translation** is the synthesis of a polypeptide, using information in the mRNA
  - **Ribosomes** are the sites of translation

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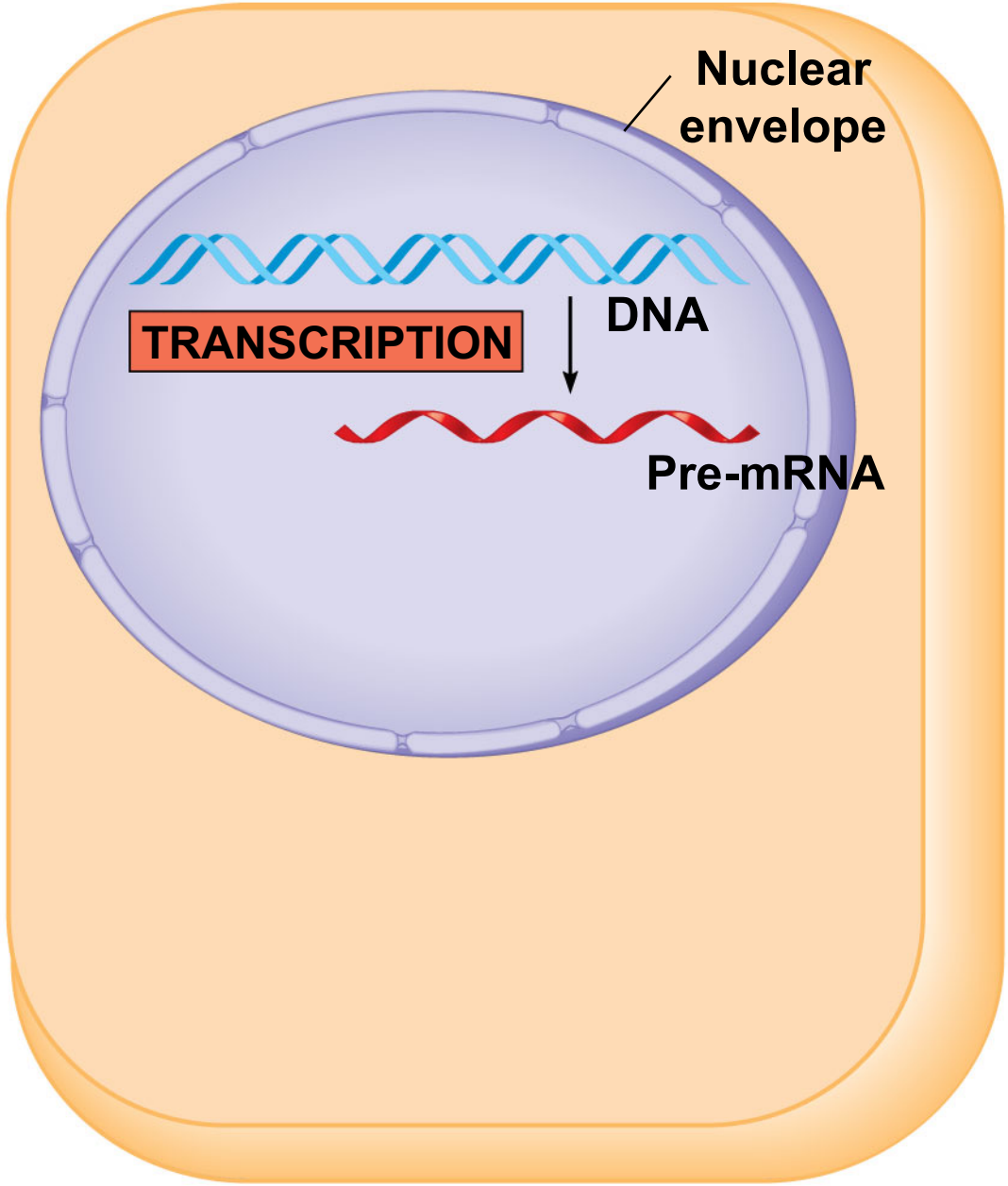


(b) Eukaryotic cell

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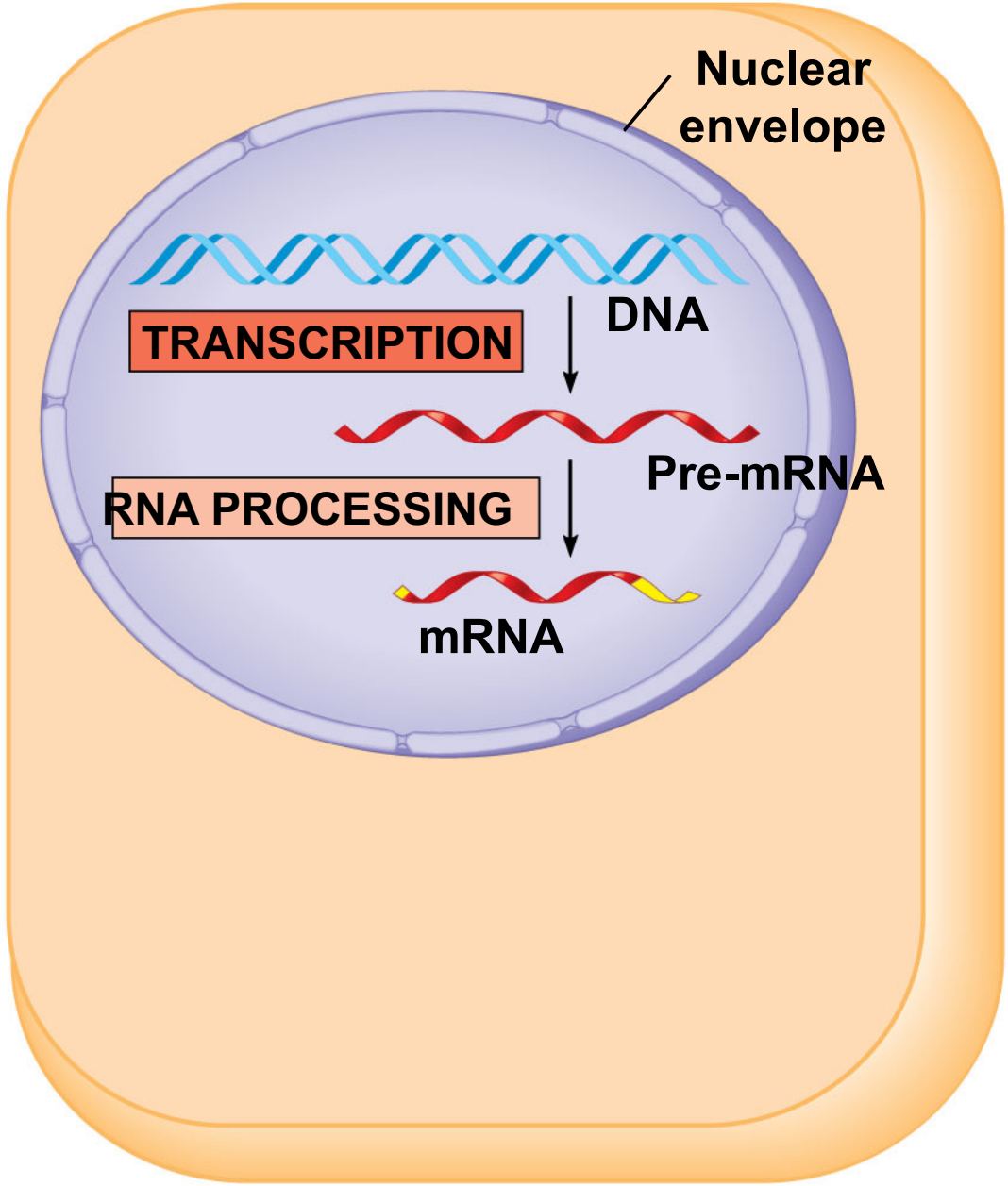


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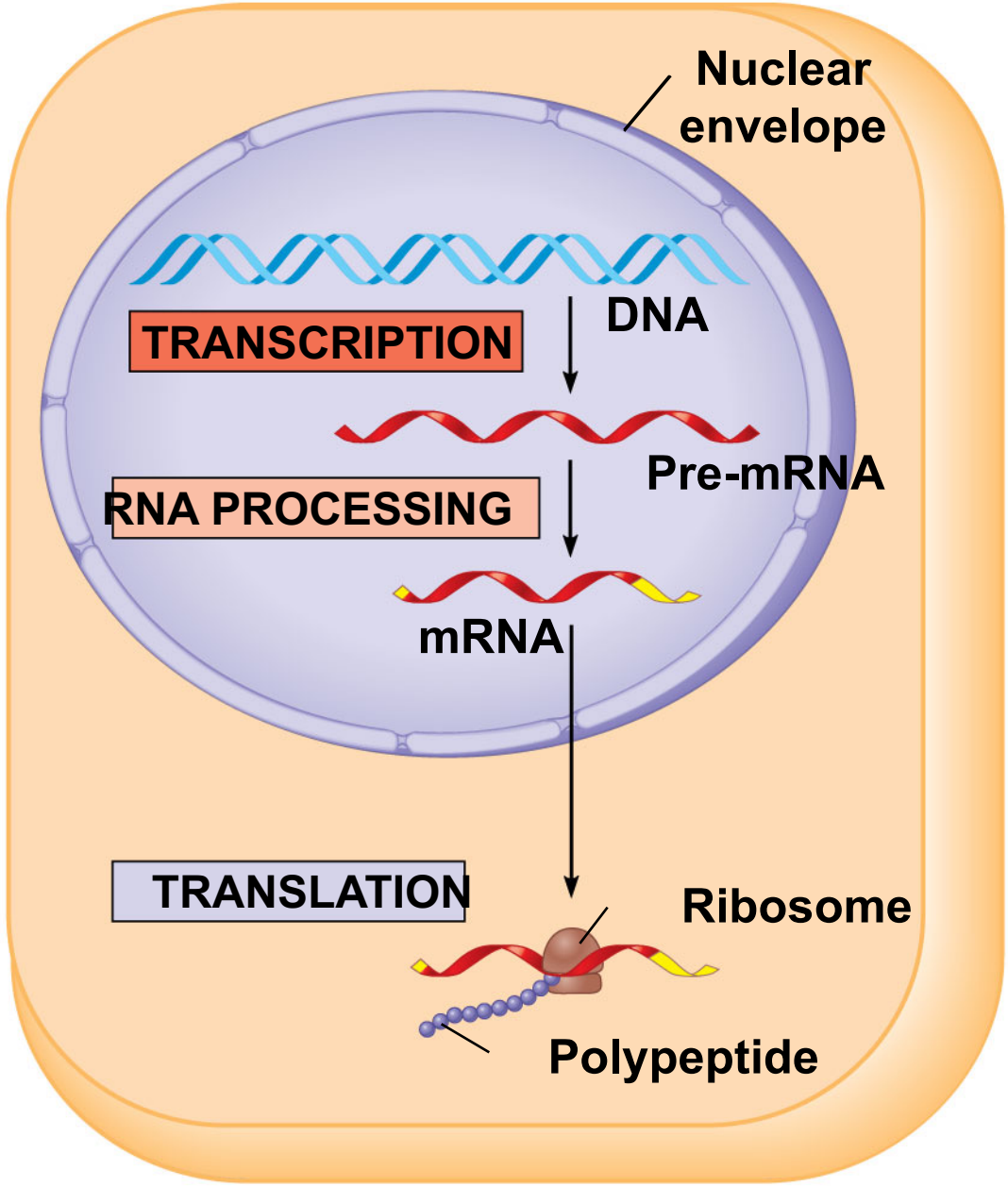


(b) Eukaryotic cell

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# Transcription

- DNA sequence information is converted to mRNA in a process called transcription

Steps:

1. The double-stranded DNA molecule is partially unzipped
2. An enzyme called RNA polymerase copies the gene's nucleotides one by one into an mRNA molecule.
   - Like DNA, mRNA is made of a particular sequence of nucleotides
   - Unlike DNA, mRNA has only a single strand, and the base Thymine in DNA is replaced by Uracil in RNA
   - Very importantly, mRNA is small and can easily exit the nucleus and go to the cytoplasm, where proteins are made.

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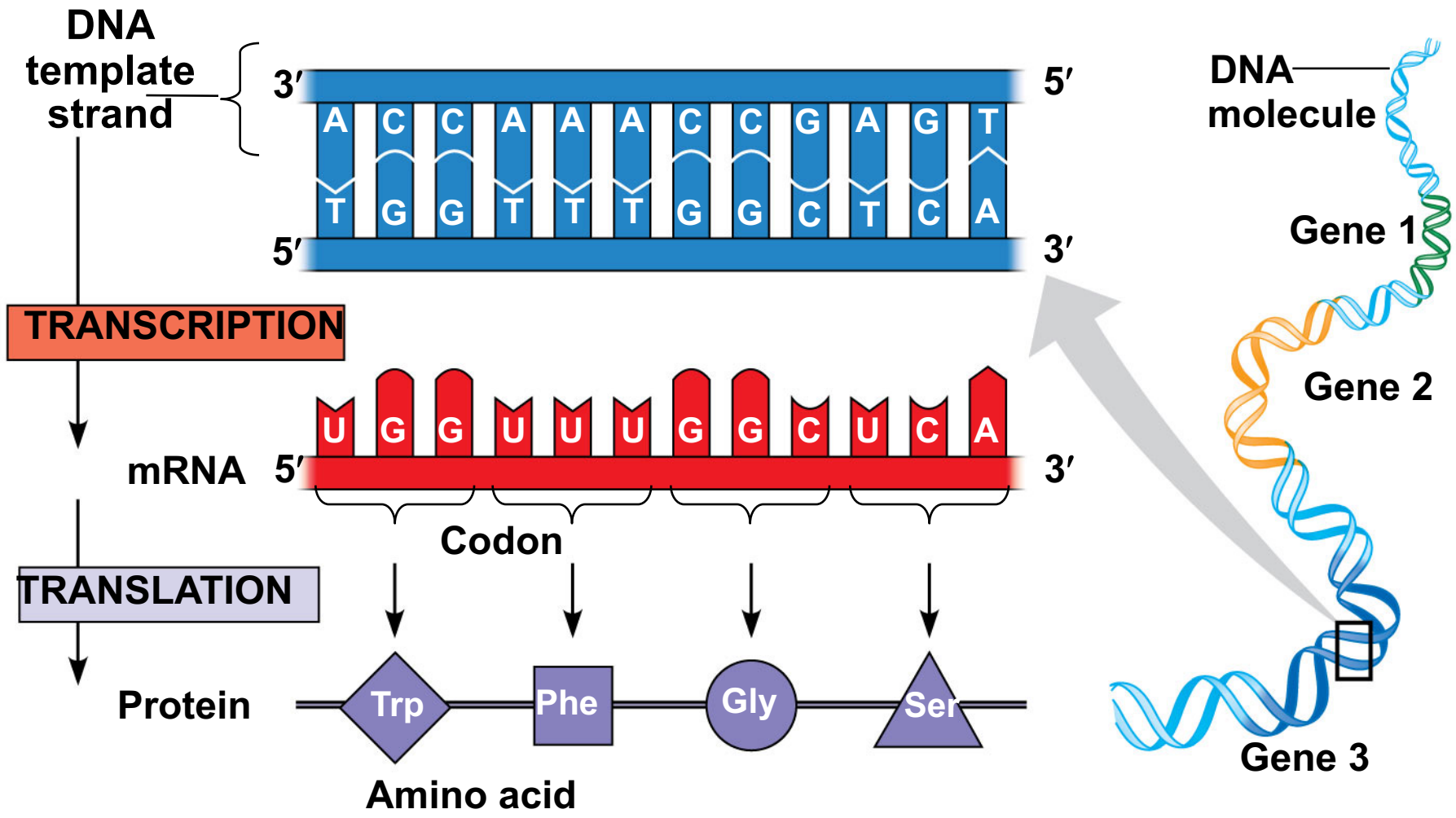
# Translation

- The sequence encoded in the mRNA molecule is decoded and converted to an amino acid sequence in a process called translation.

Steps:

1. After a mature mRNA leaves the nucleus of a cell, it heads for a ribosome in the cell's cytoplasm where the code it contains can be translated to produce a protein
2. As the strand of mRNA slides through the ribosome, the code is read three nucleotides at a time (called a codon).
3. Each codon specifies one amino acid in the polypeptide chain of a protein which are linked together to form a protein.

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https://www.youtube.com/watch?v=gG7uCskUOrA



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# Comparison between translation and transcription

| | Transcription | Translation |
|---|---|---|
| Purpose | To make RNA copies of individual genes | To synthesize proteins which are used for cellular functions |
| Definition | Transcription is the synthesis of RNA from a DNA template. | Translation is the synthesis of a protein from an mRNA template. i.e., translating the DNA genetic information carried by mRNA to synthesize proteins. This is the second step of gene expression. |
| Products | mRNA | Proteins |
| Location | Nucleus | Cytoplasm |